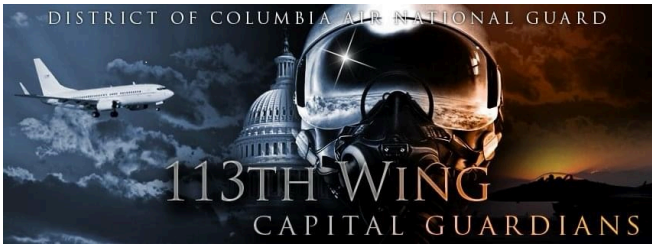
Job Safety Training Outline

Built to align with DAFI 91-202 paragraph 14.1 requirements.



Unit: 113 Maintenance Squadron

Work Center: AIS

Work Area / Office Symbol: CMF/MXMVT

Phone: 312-857-6298

Supervisor: Michael Touchette, MSgt, Avionics Back Shop Supervisor

Reviewer: Scott Morrill

Review Date: 2026-05-17

Effective Date: 2026-05-19

Work Center Overview

The Avionics Intermediate Shop (AFSC: 2A0X1) Personnel perform and manage avionics test stationfunctions and activities for F-16 aircraft. Personnel operate, inspect, maintain, program, and calibratecomputer and manually operated avionics test equipment, support equipment, and aircraft avionicssystems components. Certain components that are tested emit radio frequency radiation. Electricalhazards include high voltage and high amperage. Repair actions include soldering, utilization of hazmatfor cleaning and adhesive purposes. Compressed Air and Compressed Nitrogen are used for cleaningand testing purposes. Personnel also process supplies through the supply system. This requires

loadingand unloading wooden crates and reusable cardboard containers of often heavy weight. Mobility requirements involve pallet building.

Work Area Hazard Analysis / JHA Notes

- **RF emitter maintenance:** Primary hazards include radiofrequency radiation, hazardous noise, heat/cold stress, and potential exposure to ethyl benzene and crystalline silica. Required controls include EMF warning signs and annual EMF, hearing-protection, and thermal-stress training; use nitrile gloves with a breakthrough time greater than four hours (over 11 mil thickness), and replace damaged or degraded gloves. Use both earmuffs (NRR 29) and disposable foam earplugs (NRR 19.1) as required; the process may occur on the flightline with engines running and warning barriers in place.

- **Soldering:** Hazards include formaldehyde, lead, silver, and tin. Personnel require initial and annual formaldehyde and lead training, initial and updated Hazard Communication training, and heat-resistant gloves; these gloves are not chemical-protective and must be replaced if damaged or chemically saturated.

- **Box packing/loading and FOD walks:** Packing and loading presents forceful-exertion hazards and requires initial and annual ergonomic training; pallet jacks, dollies, and hydraulic lifts are used for heavy parts. FOD walks expose personnel to aircraft noise and heat/cold stress; hearing and thermal-stress training, earplugs, and earmuffs are required.

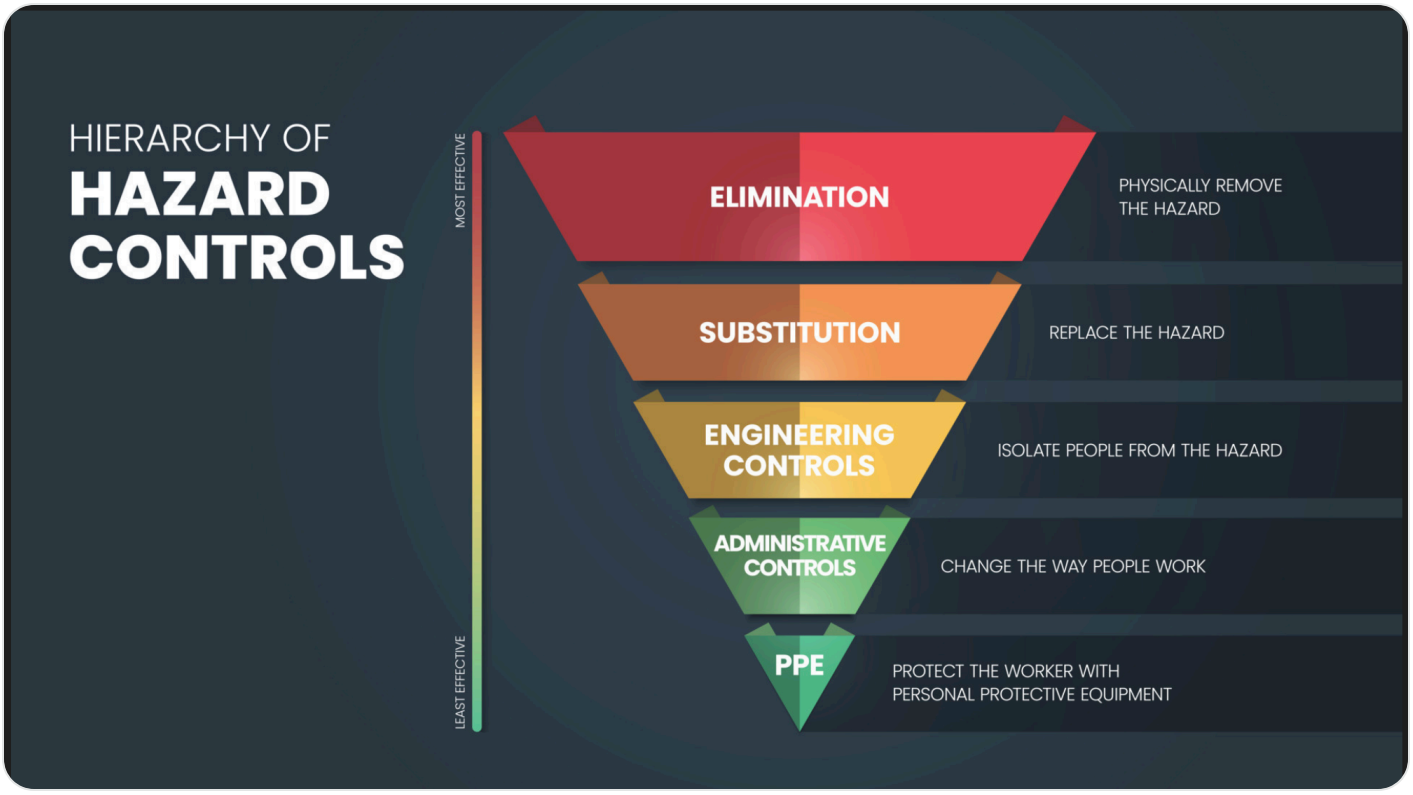
- **Special restrictions:** Document required training on AF Form 55 or equivalent, inspect, clean, maintain, and replace PPE as needed, and follow Bioenvironmental Engineering work/rest or warming-cycle guidance. Listed controls are considered adequate only when properly maintained and used; the assessment must remain posted on the health and safety board until replaced.

Required Training Topics

- **Workplace hazards:** Describe crushing, burn, chemical, ladder, weather, flight line, and noise hazards specific to the work center, plus the technical orders, JHAs, and instructions workers must follow.
- **Personal protective equipment:** Train employees on required PPE, including donning, doffing, cleaning, maintenance, storage, disposal, and reporting contact-lens or medical issues that could affect wear.
- **Emergency action and fire prevention:** Cover alarms, AEDs, extinguishers, evacuation routes, shelter locations, and response procedures tied to the work center.
- **Reporting unsafe conditions:** Instruct personnel to report unsafe equipment, conditions, procedures, and injuries without fear of reprisal, including use of DAF Form 457 and applicable reporting channels.
- **Mishap / injury / illness reporting:** Explain on-duty and off-duty reporting requirements, medical treatment procedures, and emergency response contacts.
- **Traffic safety:** Brief seat belt, speed limit, spotter, electronic device, and motorcycle safety requirements applicable to the work center.
- **DAFVA 91-209 and DAFSMS:** Review the location of DAFVA 91-209 and how workers contribute to the unit safety management system.

Hierarchy of Controls

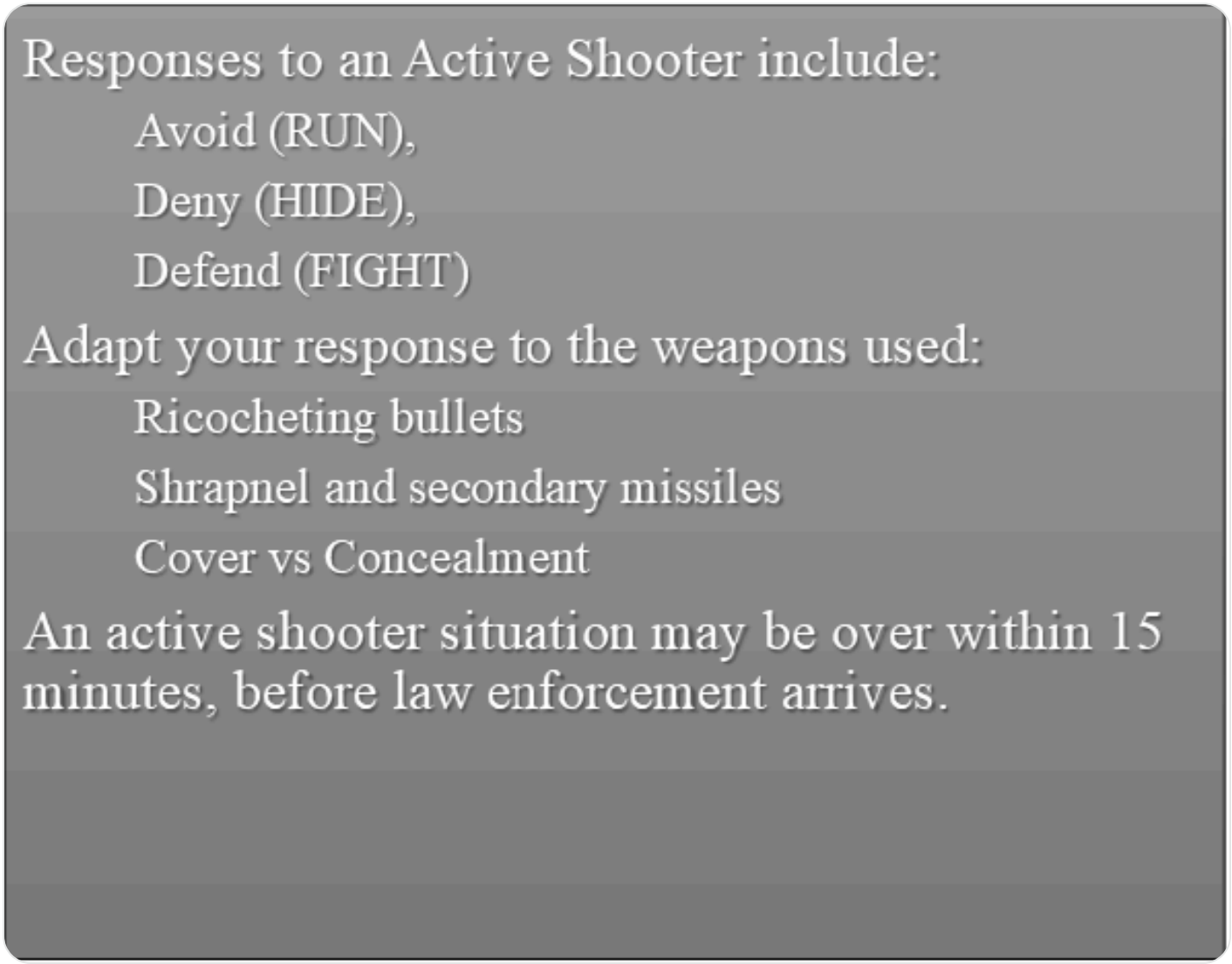
Apply these controls from most effective to least effective when reducing workplace hazards under DAFI 91-202 paragraph 14.1.2.1.4.



Use the highest level of control feasible first. PPE should protect workers only after elimination, substitution, engineering, and administrative controls have been considered.

Active Shooter Response

In the event of an active shooter, individuals should employ the **Avoid, Deny, Defend** response method. This strategy is adaptable to the specific situation and environment.



Avoid (Run): The first and preferred course of action is to evacuate the area if a safe route is available. Maintain awareness of your surroundings and have an escape route planned. Leave your belongings, evacuate regardless of whether others follow, and do not attempt to move wounded people. Once you have reached a safe location, call 911 and provide any information you have.

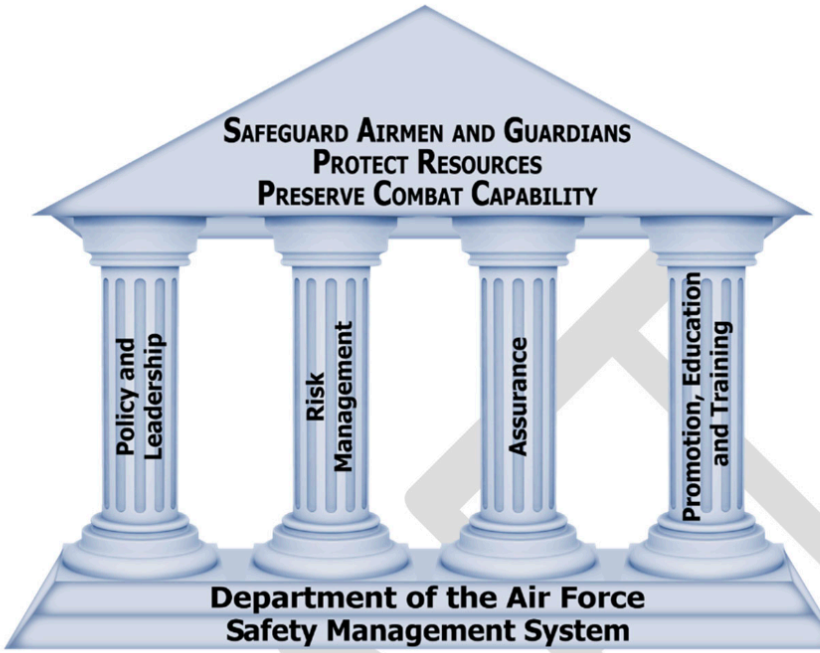
Deny (Hide): If you cannot evacuate, your next priority is to deny the attacker access to your location. This involves more than just hiding; it means securing your position. Lock and blockade doors with heavy furniture, turn off lights, silence your phone, and remain out of the shooter's view. Remember that concealment only hides you, while cover can offer protection from bullets.

Defend (Fight): As a last resort, when your life is in immediate danger, you must be prepared to defend yourself. Act aggressively, improvise weapons from your surroundings, and commit fully to your actions. Taking action may be risky, but it could be your best or only chance for survival.

When law enforcement arrives: Remain calm, follow all instructions, keep your hands visible with fingers spread, and be prepared to provide information about the shooter’s location, number, and description.

DAFSMS Framework

DAFSMS uses four pillars and the DAFSMS framework to structure the mishap prevention program through continuous improvement and the Plan-Do-Check-Act model.



Policy and Leadership

Safety policy provides the structure for a proactive mishap prevention program. Active leadership involvement at every level is critical to implementation and sustained execution.

Risk Management

Integrate risk management into all safety-related activities so hazards are identified early, controls are implemented, and the safety culture stays proactive.

Assurance

Use evaluations, monitoring, and review processes to measure conformance to standards and drive continuous improvement across the mishap prevention program.

Promotion, Education, and Training

Provide safety awareness, embed ongoing training, implement effective controls, and keep airmen and guardians actively engaged in mishap prevention.

Department of the Air Force SMS Framework

Policy and Leadership

- Leadership planning, organization, direction, and control

- Management roles, responsibilities, and relationships
- Procedures and controls that are developed, documented, maintained, and monitored
- Safety and quality balance, integration, and goal setting

Risk Management

- Hazard identification through mission/task analysis and hazard lists
- Hazard assessment using exposure, probability, severity, and risk level
- Control options, effects, prioritization, selection, and decisions
- Implementation, communication, accountability, support, supervision, and feedback

Assurance

- Performance requirements, expectations, and control effectiveness
- Continuous monitoring, self-inspections, internal and external inspections
- Mishap and event investigation, reporting, metrics, and improvement opportunities
- Change management and continuous improvement actions

Promotion, Education, and Training

- Culture built on informed, flexible, learning, just, and reporting behaviors
- Competency through training and education
- Awareness through internal communication, external communication, and information access

Job Specific Training Items

The following modules apply to this work center and should be reviewed with trainees before task performance.

Hazardous Energy ControlHazard CommunicationHearing ConservationConfined Space ProgramCPR TrainingElectromagnetic Field TrainingLadder SafetyActive Shooter ResponseGovernment Vehicle / Utility Vehicle UseManual Lifting and Material HandlingFire Protection and PreventionErgonomics / Office and Administrative Workstation SafetyJewelry / Loose Articles RestrictionsThermal Stress / Heat and Cold Stress

- **Hazardous Energy Control:** DAFMAN 91-203 establishes a formal Hazardous Energy Control Program with dedicated training, documented procedures, and periodic self-inspections before servicing or maintenance work is performed. Train authorized employees on hazardous energy sources, magnitude, and isolation methods; affected employees on the purpose and use of procedures; and other nearby employees on restart/reenergization prohibitions. Retrain when assignments, equipment, processes, or procedures change, or when inspections show gaps. DAFMAN 91-203 - Air Force Occupational Safety, Fire and Health Standards Chapter 21; 29 CFR 1910.147 - The control of hazardous energy (lockout/tagout)
- **Hazard Communication:** Employees potentially exposed to chemical hazards must complete HAZCOM training upon initial assignment and whenever a new hazard or chemical is introduced or new tasks create new hazards. Supervisors must ensure SDS access for chemicals used in the

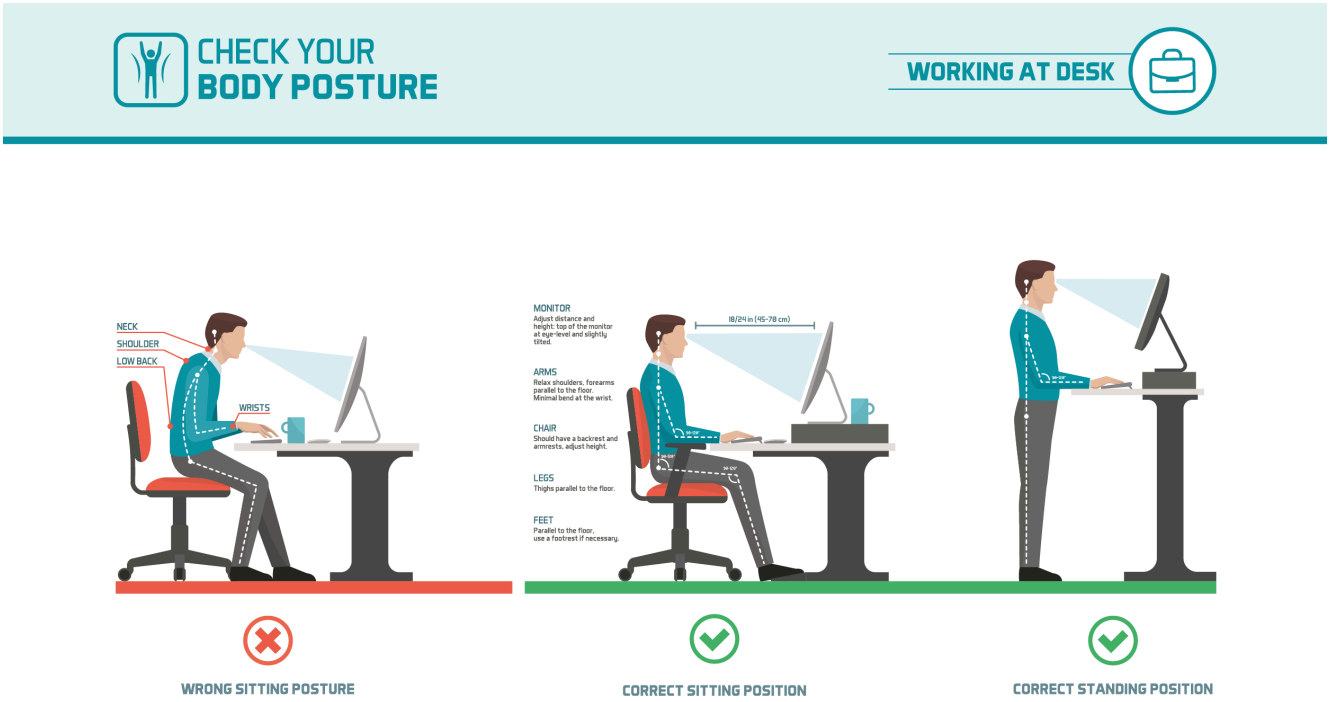
work process. Provide effective training at initial assignment and whenever a new chemical hazard is introduced. Cover where hazardous chemicals are present, the written program, labels and SDS access, how releases are detected, hazard types, protective measures, emergency procedures, and required PPE. AFI 90-821 - Hazard Communication (HAZCOM) Program; 29 CFR 1910.1200 - Hazard Communication

- **Hearing Conservation:** Hazardous noise concerns must be evaluated and controls recommended through the occupational health risk assessment process. Use the AF Hearing Conservation Program as the training and control basis when shop noise hazards are identified. Provide a hearing conservation training program for employees exposed at or above the action level, repeat it annually, and update it when equipment or work processes change. Cover hearing effects, protector selection and care, and the purpose of audiometric testing. AFI 48-127 - Occupational Noise and Hearing Conservation Program
- **Confined Space Program:** Employees entering confined spaces must be trained to safely accomplish assigned duties and rescue procedures for the specific permit-required space entered. Safety observers and attendants must also be trained on rescue procedures for each type of permit-required confined space. Train affected employees before first assignment, before duty changes, when permit-space operations introduce new hazards, and when deviations or inadequate knowledge are identified. Training must establish proficiency and be documented by employee, trainer, and date. DAFMAN 91-203 - Air Force Occupational Safety, Fire and Health Standards Chapter 23; 29 CFR 1910.146 - Permit-required confined spaces
  Link to training: https://usaf.percipio.com/linked-contents/98e905dc-f566-4212-9bbc-9bc72499def6/landing
- **CPR Training:** Initial first aid and CPR training must be completed before assigning duties that require it, refresher training must occur before certification expires, and CPR training must include PAD instruction. Remote or high-risk work must have an immediate medical response plan. Where work conditions require trained first-aid responders, ensure personnel are adequately trained and available. OSHA 2254 links first-aid and CPR availability to certain work settings, but local mission risk and AF requirements should define who must hold current CPR credentials. DAFMAN 91-203 - Air Force Occupational Safety, Fire and Health Standards Chapter 1
  Link to training: https://www.redcrosslearningcenter.org/s/online-class-registration?Id=a65Vx000007CYxxIAG
- **Electromagnetic Field Training:** Employees with implanted medical devices must notify their supervisors before work and seek medical assessment before duties in EMF environments. Use AF EMF program controls and local work restrictions as the training basis. Use AF EMF program guidance for source identification, posted restrictions, exposure boundaries, and work controls. The closest OSHA 2254 crosswalk is electrical safety training for employees exposed to shock or electrical hazards and the electric power standard’s job-assignment-specific safety practices and emergency procedures. AFI 48-109 - Electromagnetic Field Radiation (EMFR) Occupational and Environmental Health Program
- **Ladder Safety:** Personnel using ladders at any working height must be trained when first assigned, and that training must include hands-on instruction covering ladder defects, electrocution hazards, positioning, and placement for various job sites. Training must be documented under DAFI 91-202. Use local ladder lesson plans, manufacturer instructions, and AF walking-working surface guidance. OSHA 2254 (2015 edition) does not provide a dedicated general-industry ladder training summary in the same way it does for lockout, forklifts, or confined spaces, so this remains primarily AF/local-driven. Common local training item often broken out separately with fixed and portable ladder rules
  Link to training: https://usaf.percipio.com/linked-contents/202f2d2b-31ce-4409-bc86-3c48f1bb51d5/landing
- **Active Shooter Response:** Tie this to the local emergency action plan, shelter/lockdown locations, notification methods, and commander-directed response procedures. Local emergency response briefing and lockdown procedures
- **Government Vehicle / Utility Vehicle Use:** Document local licensing, route restrictions, spotter use, rollover protection, and unit-specific operating rules. OSHA 2254 does not provide a direct

GOV/UTV training excerpt here. Local GOV, GVO, or UTV operating rules and licensing expectations

- **Manual Lifting and Material Handling:** Supervisors must ensure thorough training on proper manual lifting techniques, required PPE, and the use of manual lifting devices. Training must use both verbal and written materials, include practice on proper motions, and be documented under DAFI 91-202. Cover shop-specific lifting limits, two-person lift triggers, hand truck and lifting-device use, and when supervisors require assisted handling. No direct OSHA 2254 standalone manual-lifting module is mapped here. Manual lifting techniques and available lifting devices
- **Fire Protection and Prevention:** Facility managers and supervisors must establish and maintain a fire prevention training program through the JSTO so employees understand their fire prevention and protection responsibilities in their work areas. This is fulfilled through job safety training and documentation under DAFI 91-202. Review the fire prevention plan with employees when the plan is developed, when responsibilities change, and when the plan changes. If extinguishers are provided for employee use, train employees on the general principles of extinguisher use and the hazards of incipient-stage firefighting at initial assignment and at least annually. Train designated users on the specific equipment they are expected to use. Local extinguisher, pull station, and fire reporting procedures
- **Ergonomics / Office and Administrative Workstation Safety:** Use DAFMAN 91-203 office safety and ergonomics guidance to brief administrative and computer-based workers on musculoskeletal disorder risk factors, workstation adjustment, posture, repetitive motion controls, manual handling of office materials, and supervisor follow-up for ergonomic concerns. Cover workstation setup, chair adjustment, monitor height and distance, keyboard and mouse position, neutral posture, repetitive motion, micro-pauses or task variation, safe lifting of office supplies, file cabinet and storage safety, early reporting of discomfort, and how to request a BE ergonomic assessment when needed. DAFMAN 91-203 - Air Force Occupational Safety, Fire and Health Standards; OSHA ergonomics guidance; local Bioenvironmental Engineering or Occupational Health guidance
- **Jewelry / Loose Articles Restrictions:** Use local shop rules and hazard analysis to define where jewelry, loose clothing, or unsecured items are prohibited around moving parts, climbing, electrical work, or snag hazards. Workplace-specific limits on rings, necklaces, watches, and loose items
- **Thermal Stress / Heat and Cold Stress:** Use the occupational health risk assessment, local BE recommendations, and supervisor hazard controls to brief personnel on thermal stress hazards and controls for hot, cold, humid, outdoor, flight line, industrial, or PPE-intensive work areas. Cover heat stress, cold stress, WBGT or locally directed exposure monitoring, acclimatization, hydration, work/rest cycles, buddy checks, PPE or clothing considerations, symptom recognition, supervisor notification, and emergency response actions for suspected heat illness or cold injury. DAFMAN 91-203 - Air Force Occupational Safety, Fire and Health Standards; DAFI 48-145 - Occupational and Environmental Health Program; local Bioenvironmental Engineering or Occupational Health guidance

Ergonomics Workstation Posture Guide:



Use this guide while briefing office and administrative workstation setup.

References

dafi48-109, dafi48-109_113MXGSUP, T.O. 00-25-234, T.O. 00-25-259, T.O. 1-1A-15, T.O. 00-25-223, T.O. 1-1-312
DAFI 48-145

Required Documents

- CA-10
  What a Federal Employee Should Do When Injured at Work. Use this as the workplace reference for employee injury reporting and follow-up actions.
- DAF Form 457
  Hazard Report. Use this form to identify, document, and elevate unsafe conditions, equipment, or procedures in the workplace.
- DAFVA 91-209 **UNITS MUST FILL OUT THEIR OWN LOCATION POSTED**
  Department of the Air Force Occupational Safety and Health Program visual aid. Keep this available so personnel know the core safety rights, responsibilities, and reporting expectations.
  Location Posted: AIS front office, Bldg. 3109 Room 130, on the shop safety board.
- LS-201
  Notice of Employee's Injury or Death (Non-Appropriated Funds). Use this when applicable for NAF employee injury or death reporting.

Emergency Information

- **Emergency Numbers:** Dial 911 which will reach Prince Georges County Police.
  Dial 301-98-9911 to reach JB Andrews Security Forces
- **Medical Facility / Treatment:** Malcolm Grow Medical Clinics and Surgery Center
  1060 W Perimeter Rd, Joint Base Andrews, MD 20762

(888) 999-1212

- **Evacuation / Muster Point:** Primary: AIS room 131 exit thru the NW double door

  Alternate: AIS room 130 exit thru the W double door

  Assembly point: Cross the rear parking lot to the North and assemble at the Gazebo
- **Shelter in Place:** In the event of a Shelter in Place announcement, personnel are directed to assemble in Building 3109 Men's Locker Room (Rm. 112)
- **Active Shooter Response Methods:** Avoid, Deny, & Defend!

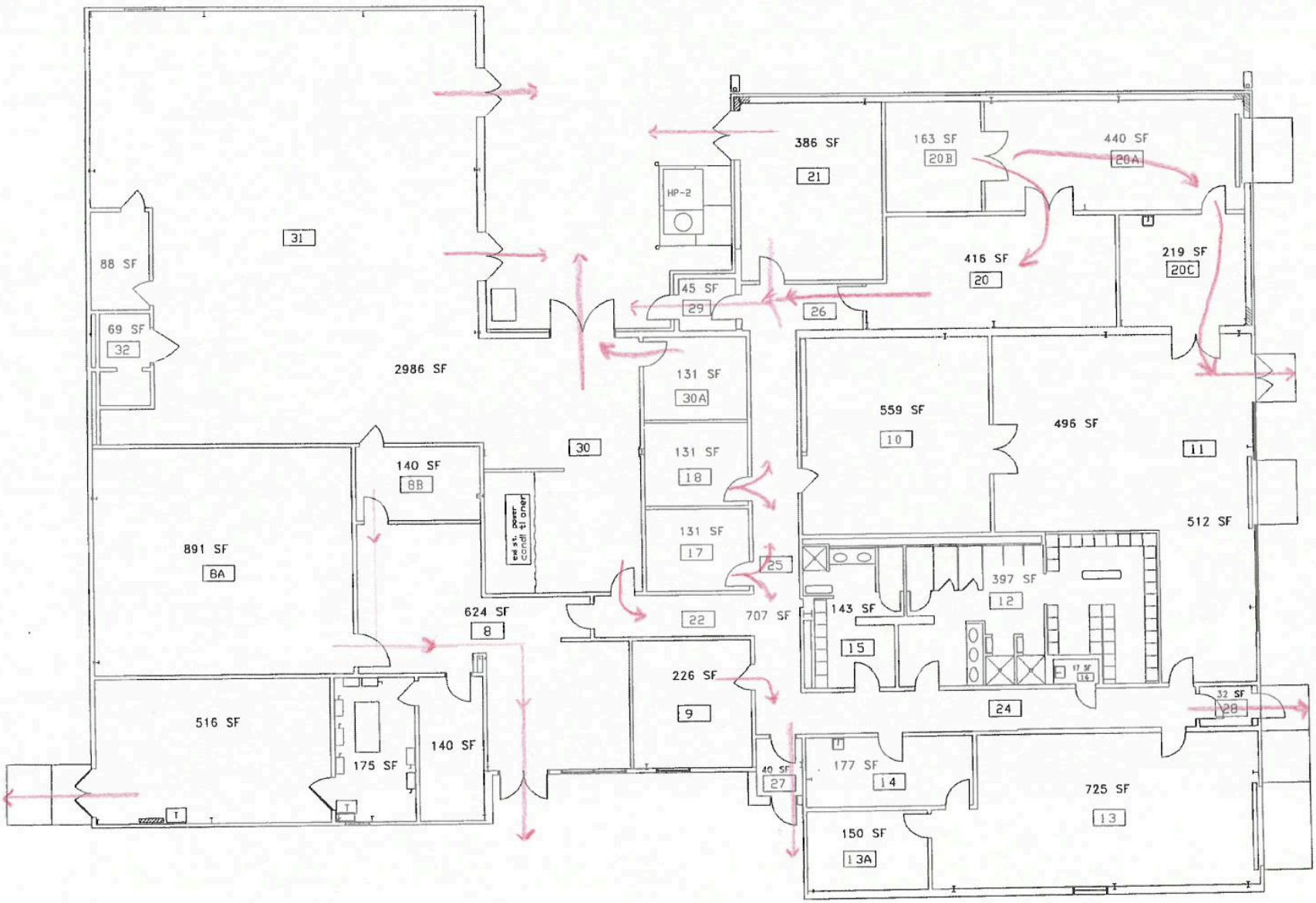
  Mass email notification the message will be "Active Shooter on JBA/Base Lockdown - TAKE COVER" followed by the last known threat area when confirmed
  – Internal PA System within some facilities
  – Follow instructions, take immediate actions
  Giant Voice
  “Lockdown,Lockdown,Lockdown.”
  All base personnel should implement base lockdown procedures
  Things to keep in mind:
  Secure your building exterior doors immediately; No one in, No one out!
  • Be aware of your surroundings and any possible dangers
  • Take note of the two nearest exits in any facility you visit
  • Leave an area if you feel uncomfortable or something does not seem right
- **Adverse Weather Shelter:** In the event of Tornado, Lightning, Hail, Flood, or severe weather; direction will be provided via:

  - Giant Voice announcements
  - Ad Hoc notifications
  - Hand held radio and phone

  If directed to Shelter in Place, personnel will gather in Building 3109, Men's Locker Room (Rm. 112)

Emergency Evacuation Route Image:

BLDG 3109 FIRE EVACUATION



Bldg 3109 Fire Escape Map.png

Emergency Equipment and Safety Board

Safety Bulletin Board Location: AIS room 130 on the East Wall

List fire pull stations, extinguishers, power cutoffs, eyewash stations, AEDs, foam systems, and any local fire/emergency equipment notes.

Emergency Equipment Maps / AED & Fire Extinguisher Location Photos:

AIS Fire Extinguisher AED Locations.pptx

Traffic Safety Program

DAF Traffic Safety Program (DAFI 91-207): Train personnel on mandatory use of seat belts and helmets, speed limits, local traffic hazards, spotters while backing, vehicle training requirements, and restrictions on electronic device use while operating vehicles on- or off-base.

Seat Belts / Occupant Restraints: All persons operating or riding in motor vehicles must wear installed restraint systems as prescribed by the manufacturer. Drivers are responsible for ensuring all passengers are briefed on and use required restraints.

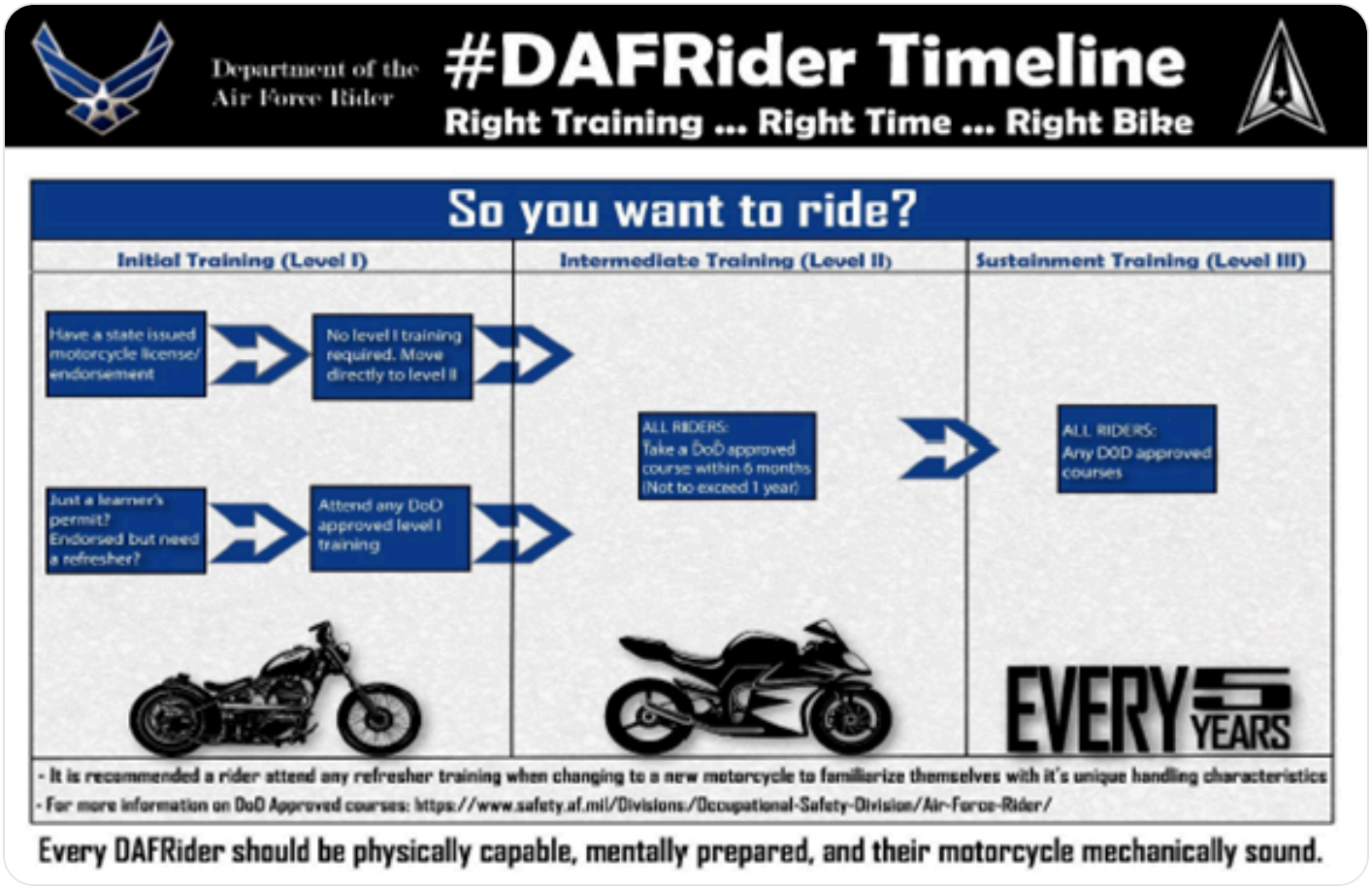
Motorcycle / ATV Training: Personnel must complete required safety training before riding a motorcycle, and local units should address required PPE, licensing, refresher training, and safety representative contacts.

Pedestrian / Bicycle Awareness: Emphasize local roadway crossing hazards, visibility requirements, spotter use, and other pedestrian or bicycle precautions specific to the installation.

Local Traffic Hazards / Vehicle Notes:
Main Gate: 24 / 7
-Virginia Gate: 0500 - 2100
-Pearl Harbor : M-F
0600-0900 (Inbound)
& 1500-1800(Out)

On Base Cautions:
Seatbelts, Pedestrians, Joggers, Cyclists, Explosive Transport Routes, Restricted Areas, and Ramp Freezes

Motorcycle Safety Representatives / Traffic Contacts:
Andrews Motorcycle Requirements
Go the the Wing Safety Office Bldg 3252 (240-857-5039)
Safety personnel will help you set up a MUSTT account and explain necessary training to ride on the installation.



Risk Assessment for Routine Use of GVOs in Traffic, Industrial, or Pedestrian Environments:

Andrews Local Traffic Awareness.pptx

Andrews Local Traffic Awareness.pptx

Reporting Unsafe Equipment, Conditions, or Procedures

Notify the supervisor immediately when unsafe equipment, conditions, or procedures are identified. Personnel may report hazards without fear of retaliation, coercion, discrimination, or reprisal.

- **DAF Form 457:** Use to document and elevate unsafe conditions, equipment, or processes.
- **SAFEREP:** Use the approved mishap or hazard reporting channel for additional reporting when directed by local procedures.

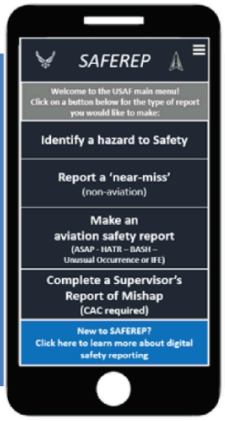
Safety Reporting … All In One Place

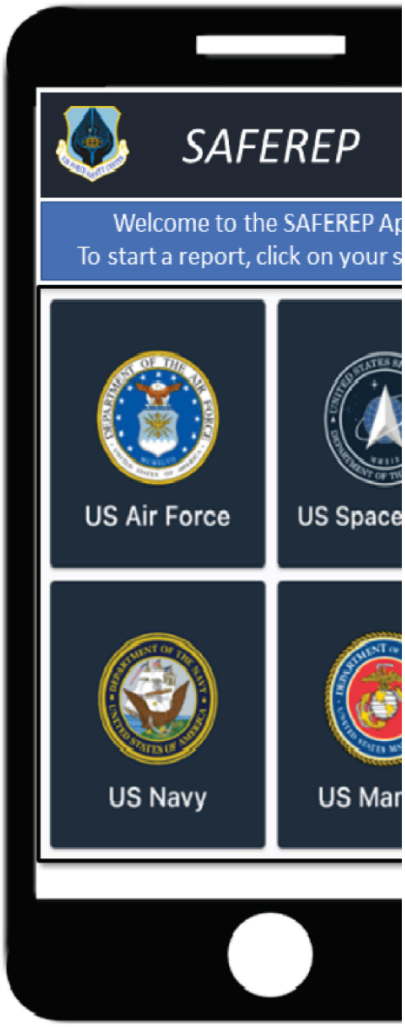
SAFEREP APP!

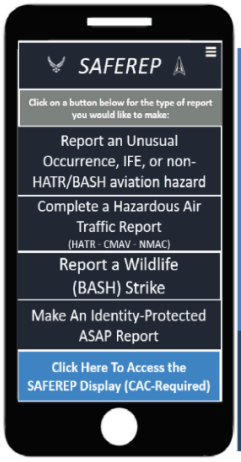
Why Safety Reporting?

Quickly and easily tell safety about safety issues you see

- In the dorms
- At work
- Anywhere on base
- In flight
- While maintaining aircraft
- On the ramp







What to report?

- Dangerous conditions
- "Near miss" events
- Aviation specific reporting
- Cultural issues that could lead to an accident, damage, or injury
- Supervisor mishap report

Report anonymously
Reports go direct to the safety office

New name, new website, new app, and a whole new look

SAFEREP is a web- and app-based way for members from every career field to identify hazards and report near-misses and even some mishaps directly to their safety offices using any computer or mobile device.

SAFEREP

Download to your mobile device from your preferred App Store

Or

Go to https://saferep.safety.af.mil









Documentation and Review

Document initial, refresher, and task-specific training in the work center record, review annually or when work conditions change, and maintain records per local records management requirements.

Bioenvironmental Workplace Survey:

DEPARTMENT OF THE AIR FORCE
ANDREWS MEDICAL GROUP (AFMEDCOM)

15 May 2026

MEMORANDUM FOR 113 MXS/MXMVT
ATTN: SCOTT MORRILL

FROM: ANDREWS OMRS/SGXB

SUBJECT: Comprehensive Health Risk Assessment, 113 Avionics Intermediate Shop
WPID:142A, Bldg 3109

1. **Introduction:** On 08 May 2026, SrA Evelin Castillo from Bioenvironmental Engineering (BE) conducted a Health Risk Assessment (HRA) of your workplace IAW DAFI 48-145, *Occupational and Environmental Health Program*, and DAFMAN 48-146, *Occupational and Environmental Health Management*. **Comprehensive HRAs are now conducted every 3 years. However, all "high risk" processes and programs will be evaluated annually.** Your shop currently does not contain high risk processes. Scott Morrill was our point of contact.

2. **Purpose:** BE conducted this assessment to review all processes for potential hazards and to identify any special surveillance required to better characterize workplace hazards and controls. The Occupational and Environmental Health Routine Letter Attachment (**Attachment 1**) briefly summarizes what your shop does and lists all identified processes (tasks), their applicable hazards, and the required controls. Any deficiencies and significant exposures are also identified on this attachment.

3. **Deficiencies:** Please note, all deficiencies found in your workplace were sent to your Squadron Commander (or equivalent) in an executive summary. Currently, there are no deficiencies assigned to your unit. BE allows 60 days to resolve occupational health deficiencies. If deficiencies are not resolved within this timeframe, BE will no longer track administrative deficiencies. IAW DAFI 48-145 para 4.3.4, "Unit commanders, workplace supervisors, facility managers and mission owners shall monitor, report and track any identified deficiencies until closure".

4. **Summary of Changes:** The following are the changes identified during this assessment.

- Added 010 Removal/Installation of aircraft components as a process.
- Added 020 Cleaning and inspection of aircraft components as a process
- Added 030 Radio Frequency (RF) Emitter Maintenance as a process
- Added 050 Soldering
- Added Lead, Tin, Silver and Formaldehyde as a hazard during the process of 050 Soldering.

6. **Conclusion:** All workers must be informed of the contents of this report and be able to view this report at any time upon request. The information contained in this report should be posted on your safety bulletin board and briefed to all employees. Employees have the right to view

their individual exposure records. Any questions may be addressed to the BE Flight at 240-857-3380.

DE TATA.TYLER.STEVEN.1590495111
Digitally signed by DE TATA.TYLER.STEVEN.1590495111
Date: 2026.05.22 12:33:27 -04'00'

TYLER S. DE TATA, SSgt, USAF
NCOIC, Occupational Health

Attachment:
Occupational and Environmental Health Routine Letter

cc:
113 MXS/CC (karl.wilson.2@us.af.mil)
113 MXEC/MXMVT (scott.morrill@us.af.mil)
113 SEG/WG (Jose.otero.4@us.af.mil)
113 AMDS/SGXB (Robert.smith.320@us.af.mil)
ANDREWS OMRS/SGXB (caesar.r.pereira.mil@health.mil)
ANDREWS OMRS/SGXB (Anthony.c.holden2.mil@health.mil)
ANDREWS OMRS/SGXB (tyler.s.detata.mil@health.mil)
ANDREWS OMRS/SGXM (denise.r.rious.civ@health.mil

Occupational and Environmental Health

Routine Letter Attachment

Date Printed: 22 MAY 2026

CUI

Installation: Andrews AFB (DRU)

Workplace: 142A 113th Avionics Intermediate Shop (113 MAINTENANCE SQ FFL8W0, MD)

Risk Category: 2 - Medium Risk

Workplace Description	# of Personnel
The Avionics Intermediate Shop (AFSC: 2A0X1) Personnel perform and manage avionics test station functions and activities for F-16 aircraft. Personnel operate, inspect, maintain, program, and calibrate computer and manually operated avionics test equipment, support equipment, and aircraft avionics systems components. Certain components that are tested emit radio frequency radiation. Personnel also process supplies through the supply system. This requires packing and unpacking wooden crates and reusable cardboard containers. POC(s): MSgt Scott Morrill Office Symbol: 113 MXS/MXMVT Duty Phone: 240-857-6298 EESOH-MIS code: 180HM Sq/CC: Lt Col Karl Wilson Sq/CCC: SMSgt Stelios Zissimos CSS number: 240-857-2785	5

Similar Exposure Groups (SEG) Associated with Workplace	142A Avionics Intermediate Shop

142A Avionics Intermediate Shop	# of Personnel
The Avionics Intermediate Shop (AFSC: 2A0X1) Personnel perform and manage avionics test station functions and activities for F-16 aircraft. Personnel operate, inspect, maintain, program, and calibrate computer and manually operated avionics test equipment, support equipment, and aircraft avionics systems components. Certain components that are tested emit radio frequency radiation. Personnel also process supplies through the supply system. This requires packing and unpacking wooden crates and reusable cardboard containers. POC(s): MSgt Scott Morrill Office Symbol: 113 MXS/MXMVT Duty Phone: 240-857-6298 EESOH-MIS code: 180HM Sq/CC: Lt Col Karl Wilson Sq/CCC: SMSgt Stelios Zissimos CSS number: 240-857-2785	5

Significant Exposures Table

Hazard Name	Process Name	OEL	Exposure Level	Need More Data	SEG Association
FORMALDEHYDE (Inhalation) (OSHA exp std) (TLV-A2: Suspected Human Carcinogen)	050 Soldering	0.02 mg/m3 (8 hr TWA)		No	142A Avionics Intermediate Shop
Rationale: IAW DAFMAN 48-146 table 3.1 – 3.3, the Health Effects Rating is high due to chronic irreversible health effects of concern The Exposure Rating is moderate because the exposure is frequently less than action level or 50% of OEL and 10 % of OEL. The Uncertainty Rating is 4 due to the level of confidence in hazard characterization and evaluated controls. IAW DAFMAN 48-146 attachment 7 criteria, confidence in Occupational Health Exposure characterization is low because quantitative data does not exist, or is insufficient to draw a conclusion regarding exposure. IAW DAFMAN 48-146 attachment 8 criteria, confidence in controls is medium because PPE is required to control exposure and workers have been observed using required PPE effectively.					
SILVER METAL (Skin and/or Eye Contact)	050 Soldering			No	142A Avionics Intermediate Shop
Rationale: The exposure evaluation is acceptable. Silver Metal is used during50 Soldering. This process is conducted weekly for a total duration of 2-4 hours. Nitrile rubber gloves are used to prevent exposure to Silver Metal. Gloves must be replaced when damaged, and upon completion of singular task requiring glove use. IAW DAFMAN 48-146 table 3.1 – 3.3, the Health Effects Rating is low due to incidental skin irritation. The Exposure Rating is moderate because the exposure is frequently less than action level or 50% of OEL and 10 % of OEL. The Uncertainty Rating is 3 due to the level of confidence in hazard characterization and evaluated controls. IAW DAFMAN 48-146 attachment 7 criteria, confidence in Occupational Health Exposure characterization is low because quantitative data does not exist, or is insufficient to draw a conclusion regarding exposure. IAW DAFMAN 48-146 attachment 8 criteria, confidence in controls is medium because PPE is required to control exposure and workers have been observed using required PPE effectively.					
LEAD (Skin and/or Eye Contact) (OSHA exp std) (2B (IARC), A3 (ACGIH), R (NTP))	020 Cleaning and inspections of Aircraft components			No	142A Avionics Intermediate Shop
Rationale: The exposure evaluation is acceptable. Lead is used during 020 Cleaning and inspections of Aircraft components. This process is conducted daily for a total duration of 2-4 hours. Nitrile rubber gloves are used to prevent exposure to Lead. Gloves must be replaced when damaged, and upon completion of singular task requiring glove use. IAW DAFMAN 48-146 table 3.1 – 3.3, the Health Effects Rating is low due to incidental skin irritation. The Exposure Rating is moderate because the exposure is frequently less than action level or 50% of OEL and 10 % of OEL. The Uncertainty Rating is 3 due to the level of confidence in hazard characterization and evaluated controls. IAW DAFMAN 48-146 attachment 7 criteria, confidence in Occupational Health Exposure characterization is low because quantitative data does not exist, or is insufficient to draw a conclusion regarding exposure. IAW DAFMAN 48-146 attachment 8 criteria, confidence in controls is medium because PPE is required to control exposure and workers have been observed using required PPE effectively.					

Process: 010 Removal/Installation of aircraft components	**Exposure Group(s) Associated with Process:**	142A Avionics Intermediate Shop
Process Frequency: Monthly	Process Duration: 30-60 minutes	
This process consists of the removal and installation of aircraft's components. Personnel use a variety of sealants and bonding throughout the process. Application methods also vary depending on task and T.O. requirements. Personnel routinely use squeeze tube sealants, liquid brush-on adhesives, sealing or bonding tapes to accomplish this process.		

Administrative

Control Name	Description	Hazard(s)
RFR - Warning Signs	Electromagnetic Field (EMF) Notice Sign	RADIOFREQUENCY RADIATION - EMF

Comments:	Required. A notice sign shall be posted where the electromagnetic field (EMF) levels exceed the lower tier maximum permissible exposures. A sign is located at the entrance to their workstation. [AFI 48-109]

Training - Hazard Communication	HAZCOM Training	ETHYL BENZENE; XYLENE
Comments:	Initial HAZCOM Training is conducted within the first 30 days of arrival and before working with hazardous materials. Supplemental training is required when changes in process or hazardous materials occur. Initial and supplemental training must be documented on AF Form 55 or equivalent as a separate training line items.	

Training - Other	Electromagnetic Field (EMF) Awareness Training	RADIOFREQUENCY RADIATION - EMF
Comments:	EMF Safety Awareness Training is conducted initially and annually then documented on AF Form 55 or equivalent. The training plans must include training and posting requirements at a minimum. The depth of information will be commensurate with the potential for exposure as well as the level of responsibility within the EMF safety program. For further guidance, please contact the Installation Radiation Safety Officer at 612-3380.	

PPE

Control Name	Description	Hazard(s)
Gloves - Nitrile Rubber	Nitrile Rubber Gloves	ETHYL BENZENE; XYLENE
PPE Narrative Comments:	These gloves have been assessed using the Quick Selection Guide to Chemical Protective Clothing and deemed adequate for use during this process and chemicals as long as the below breakthrough times are adhered to.	

Process: 020 Cleaning and inspections of Aircraft components	**Exposure Group(s) Associated with Process:**	142A Avionics Intermediate Shop
Process Frequency: Daily	Process Duration: 2-4 hours	
Personnel are responsible for the cleaning and preventive maintenance of aircraft components, equipment and tools routinely. This process is performed in conjunction with an inspection, repair, replacement of a damaged part, etc.		

Administrative

Control Name	Description	Hazard(s)
Training - Crystalline Silica	Silica Training	SILICA, CRYSTALLINE CRISTOBALITE; SILICA, CRYSTALLINE, QUARTZ; SILICA, CRYSTALLINE, TOTAL RESPIRABLE
Comments:	rystalline silica training is provided initially and annually then documented on AF Form 55 or equivalent. Crystalline Silica training must meet all the requirements in OSHA 29 CFR 1910.1053. Please contact Public Health for the training material.	

Training - Hazard Communication	HAZCOM Training	ETHYL BENZENE; SILICA, CRYSTALLINE CRISTOBALITE; SILICA, CRYSTALLINE, QUARTZ; SILICA, CRYSTALLINE, TOTAL RESPIRABLE
Comments:	Initial HAZCOM Training is conducted within the first 30 days of arrival and before working with hazardous materials. Supplemental training is required when changes in process or hazardous materials occur. Initial and supplemental training must be documented on AF Form 55 or equivalent as a separate training line items.	

PPE

Control Name	Description	Hazard(s)
Gloves - Nitrile Rubber	Nitrile Rubber Gloves	ETHYL BENZENE; SILICA, CRYSTALLINE CRISTOBALITE; SILICA, CRYSTALLINE, QUARTZ; SILICA, CRYSTALLINE, TOTAL RESPIRABLE

PPE Narrative Comments: These gloves have been assessed using the Quick Selection Guide to Chemical Protective Clothing and deemed adequate for use during this process and chemicals as long as the below breakthrough times are adhered to.

Breakthrough time: >4 hrs (Nitrile glove thickness >11 mil)

(Technician note: the most limiting breakthrough time indicated). Replace gloves if torn, cut, punctured or otherwise damaged. Thicker gloves provide better abrasion resistance and chemical resistance, but reduced dexterity. No single glove material protects against all chemical exposures. Inspect for dry rot.

Process: 030 Radio Frequency (RF) Emitter Maintenance	Exposure Group(s) Associated with Process:	142A Avionics Intermediate Shop
Process Frequency: Weekly	Process Duration: 1-2 hours	

Personnel perform ops checks and maintenance on RF emitters. Engines may be running to accomplish. This process may be conducted outside on the flightline and visible warning barriers are used.

Administrative

Control Name	Description	Hazard(s)
RFR - Warning Signs	Electromagnetic Field (EMF) Notice Sign	RADIOFREQUENCY RADIATION - EMF

Comments: Required. A notice sign shall be posted where the electromagnetic field (EMF) levels exceed the lower tier maximum permissible exposures. A sign is located at the entrance to their workstation. [AFI 48-109]

Training - Hearing Protection Program	Hazardous Noise Training	NOISE

Comments: Personnel require initial & annual workplace-specific hearing protection program training from shop supervisor, however, they are not enrolled in the Hearing Conservation Program. Document training on AF Form 55 or equivalent.

Training - Heat/Cold Stress	Thermal Awareness Training	COLD STRESS; HEAT STRESS

Comments: Thermal Injury Protection Program Training is conducted initially and annually. Personnel follow work/rest cycles (or warming cycles, frost bite warnings) dictated by Bioenvironmental Engineering's WBGT readings (or local wind chill charts). Training is documented on AF Form 55 or equivalent.

Training - Other	Electromagnetic Field (EMF) Awareness Training	RADIOFREQUENCY RADIATION - EMF

Comments: EMF Safety Awareness Training is conducted initially and annually then documented on AF Form 55 or equivalent. The training plans must include training and posting requirements at a minimum.

The depth of information will be commensurate with the potential for exposure as well as the level of responsibility within the EMF safety program. For further guidance, please contact the Installation Radiation Safety Officer at 612-3380.

PPE

Control Name	Description	Hazard(s)
Hearing - Ear/Circumaural Muff	Golden Eagle Tasco Ear Muffs	NOISE
PPE Narrative Comments:	Replace ear muffs when headbands are so stretched that they do not keep ear cushions snuggly against the head. Replace ear cushions that are no longer flexible. NRR 29	

Hearing - Earplug, Disposable, Foam Plastic	Foam Ear Plugs - SoundGuard Two-Color	NOISE
PPE Narrative Comments:	Replace disposable ear plugs when they no longer form an airtight seal when properly inserted. (ANSI 12.6-1984) NRR: 19.1	

Process: 050 Soldering	**Exposure Group(s) Associated with Process:**	142A Avionics Intermediate Shop
Process Frequency: Weekly	Process Duration: 30-60 minutes	
Personnel perform soldering to repair electrical contacts and wiring within the aircraft.	Soldering is done using a low heat solder gun at <500	

Administrative

Control Name	Description	Hazard(s)
Training - Formaldehyde	Formaldehyde Training	FORMALDEHYDE
Comments:	Formaldehyde training is provided initially and annually then documented on AF Form 55 or equivalent. Formaldehyde training must meet all the requirements in OSHA 29 CFR 1910.1048. Please contact Public Health for the training material.	

Training - Hazard Communication	HAZCOM Training	FORMALDEHYDE; LEAD; SILVER METAL; TIN
Comments:	Initial HAZCOM Training is conducted within the first 30 days of arrival and before working with hazardous materials. Supplemental training is required when changes in process or hazardous materials occur. Initial and supplemental training must be documented on AF Form 55 or equivalent as a separate training line items.	

Training - Lead (29 CFR 1910 Subpart Z - Toxic and Hazardous Substances)	Lead Training	LEAD
Comments:	Lead training is provided initially and annually then documented on AF Form 55 or equivalent. Lead training must meet all the requirements in OSHA 29 CFR 1910.1025. Please contact Public Health for the training material.	

PPE

Control Name	Description	Hazard(s)
Hand - Other	Heat Resistance Gloves	FORMALDEHYDE; LEAD; SILVER METAL; TIN
PPE Narrative Comments:	Will not protect hands if they become saturated with hazardous chemicals. High abrasion resistance and short breakthrough times. Not designed for chemical protection. Replace if torn, cut, punctured, or otherwise damaged.	

Process: Box Packing and Loading	Exposure Group(s) Associated with Process:	142A Avionics Intermediate Shop
Process Frequency: 2-3 Times/Week	Process Duration: 1-2 hours	
Personnel process supplies through the supply system. This requires packing and unpacking wooden crates and reusable cardboard containers. Large parts weigh between 100 and 200 lbs. Personnel use pallet jack, dolly, and hydraulic lift to assist with loading containers onto vehicles. This process takes about 2 hours to complete. No chemicals are introduced during this process.		

Administrative

Control Name	Description	Hazard(s)
Training - Other	Workplace Specific Ergonomic Training	Forceful Exertion
Comments:	Ergonomic Training is conducted initially and annually then documented on AF Form 55 or equivalent	

Process: Foreign Object Debris (FOD) Walk	Exposure Group(s) Associated with Process:	142A Avionics Intermediate Shop
Process Frequency: Daily	Process Duration: 15-30 minutes	
Personnel walk the flight line tarmac daily for approximately 15-20 minutes to pick up miscellaneous debris. Personnel wear earplugs and earmuffs during this process due to hazardous noise exposure from aircraft. Personnel are also exposed to thermal stress and cold stress during this process. No chemicals are introduced into the process		

Administrative

Control Name	Description	Hazard(s)
Training - Hearing Protection Program	Hazardous Noise Training	NOISE
Comments:	Personnel require initial & annual workplace-specific hearing protection program training from shop supervisor, however, they are not enrolled in the Hearing Conservation Program. Document training on AF Form 55 or equivalent.	
Training - Heat/Cold Stress	Thermal Stress Training	COLD STRESS; HEAT STRESS
Comments:	Thermal Injury Prevention Program Awareness Training - Thermal Injury Protection Program Training is conducted initially and annually. Personnel follow work/rest cycles (or warming cycles, frost bite warnings) dictated by Bioenvironmental Engineering's WBGT readings (or local wind chill charts). Training is documented on AF Form 55 or equivalent.	

PPE

Control Name	Description	Hazard(s)
Hearing - Ear/Circumaural Muff	Golden Eagle Tasco Ear Muffs	NOISE
PPE Narrative Comments:	Replace ear muffs when headbands are so stretched that they do not keep ear cushions snuggly against the head. Replace ear cushions that are no longer flexible. NRR 29	
Hearing - Earplug, Disposable, Foam Plastic	Foam Ear Plugs - SoundGuard Two-Color	NOISE

PPE Narrative Comments:	Replace disposable ear plugs when they no longer form an airtight seal when properly inserted. (ANSI 12.6-1984) NRR: 19.1

Acronyms: dBA=decibels A-weighted; mg/m3 = milligrams per cubic meter; OEEL = Occupational and Environmental Exposure Limit; ppm = parts per million; TLV = Threshold Limit Value; TWA = Time-weighted Average; WPID = Workplace Identifier Code

Boldfaced Font: A red boldfaced "health hazard" indicates OSHA Expanded Standards apply; a boldfaced "health hazard" indicates teratogens/carcinogens; a boldfaced "measured exposure" indicates personnel may be exposed above 50% of OEEL; a red boldfaced "measured exposure" indicates personnel may be exposed above OEEL.

The Routine Assessment Shop Report Table listed above for each process is mandatory unless otherwise stated. The listed controls (Admin, Engineering, and PPE) may differ from the recommended controls on the Safety Data Sheets (SDS formerly MSDS) and any concerns or questions regarding these differences should be addressed to BE for clarification. This attachment must be posted on your health and safety board until replaced by an updated listing. Reference AFI 91-203 & AFMAN 48-146, or contact your safety representative for additional requirements. Personnel must receive training in the use, cleaning, maintenance, and inspection of each PPE item before use. If maintained and used properly, the controls listed are adequate to protect workers from the identified health hazards.

Bioenvironmental Engineering Craftsman (sign and date)	Bioenvironmental Engineer or Industrial Hygienist (sign and date)
DE TATA.TYLER.STEVEN.1590495111 Digitally signed by DE TATA.TYLER.STEVEN.1590495111 Date: 2026.05.22 12:33:44 -04'00'	

CUI HLTH//TSCA

Controlled By: DHA Categories: HLTH//TSCA
LDC: FEDCON, DOD Only
POC:DE TATA, T

Applicable Job Hazard Analyses:

- 2026 OEHED 142A 113 MXS.pdf

United States Air Force

Occupational and Environmental Health Exposure Data

Reference: DAFI 48-145

Current as of Date: 22 May 2026 ***Date Printed: 22 May 2026***

(CUI)

Workplace	Risk Categorization	Installation	Organization	# of SEGs	# of Personnel
142A 113th Avionics Intermediate Shop (WPID: MXAV)	2 - Medium Risk	Andrews AFB (DRU)	113 MAINTENANCE SQ FFL8W0, MD	1	5

NOTE: This report includes occupational health risk assessment data from the Defense Occupational & Environmental Health Readiness System for Industrial Hygiene (DOEHRS-IH) database. Data is considered reviewed, approved, and current by the below certifying authority as of the above Current as of Date.

Exposure Group(s) Associated with Workplace

142A Avionics Intermediate Shop

Administrative Controls/Training

Training - (Other (Workplace Specific Ergonomic Training)); Training - (Other (Electromagnetic Field (EMF) Awareness Training)); Training - (Lead (29 CFR 1910 Subpart Z - Toxic and Hazardous Substances) (Lead Training)); Training - (Heat/Cold Stress (Thermal Stress Training)); Training - (Heat/Cold Stress (Thermal Awareness Training)); Training - (Hearing Protection Program (Hazardous Noise Training)); Training - (Hazard Communication (HAZCOM Training)); Training - (Formaldehyde (Formaldehyde Training)); Training - (Crystalline Silica (Silica Training)); RFR - (Warning Signs (Electromagnetic Field (EMF) Notice Sign))

Workplace Description

The Avionics Intermediate Shop (AFSC: 2A0X1) Personnel perform and manage avionics test station functions and activities for F-16 aircraft. Personnel operate, inspect, maintain, program, and calibrate computer and manually operated avionics test equipment, support equipment, and aircraft avionics systems components. Certain components that are tested emit radio frequency radiation. Personnel also process supplies through the supply system. This requires packing and unpacking wooden crates and reusable cardboard containers.

POC(s): MSgt Scott Morrill

Office Symbol: 113 MXS/MXMVT

Duty Phone: 240-857-6298

EESOH-MIS code: 180HM

Sq/CC: Lt Col Karl Wilson
Sq/CCC: SMSgt Stelios Zissimos
CSS number: 240-857-2785

Exposure Group: 142A Avionics Intermediate Shop **# SEG Personnel:** 5

Health Hazard	Regulatory Limit (Duration) Agency	Measured Exposure	Exposure Determination
NOISE	85 dBA (8 hr TWA) DoD	58 dBA	Below Action Level
Existing Control(s)	Hearing - Earplug, Disposable, Foam Plastic (Foam Ear Plugs - SoundGuard Two-Color); Hearing - Ear/Circumaural Muff (Golden Eagle Tasco Ear Muffs)		
Process(es)	Foreign Object Debris (FOD) Walk (Daily 15-30 minutes); 030 Radio Frequency (RF) Emitter Maintenance (Weekly 1-2 hours)		
RADIOFREQUENCY RADIATION - EMF	010 mW/cm2 (6 min Ceiling) ACGIH		
Process(es)	030 Radio Frequency (RF) Emitter Maintenance (Weekly 1-2 hours)		

Page 1 of 2

Acronyms: dBA=decibels A-weighted; mg/m3 = milligrams per cubic meter; OEEL = Occupational and Environmental Exposure Limit; ppm = parts per million; TLV = Threshold Limit Value; TWA = Time-weighted Average; WPID = Workplace Identifier Code; Cens = Measured Result Includes Censored Data
Boldfaced Font: A red boldfaced "health hazard" indicates OSHA Expanded Standards apply; a boldfaced "health hazard" indicates teratogens/ carcinogens; a boldfaced "measured exposure" and/or "exposure determination" indicates personnel may be exposed above 50% of OEEL; a red boldfaced "measured exposure" and/or "exposure determination" indicates personnel may be exposed above OEEL.

CERTIFYING AUTHORITY	
Bioenvironmental Engineering Craftsman (sign and date)	Bioenvironmental Engineer or Industrial Hygienist (sign and date)
DE TATA.TYLER.STEVEN.1590495111 Digitally signed by DE TATA.TYLER.STEVEN.1590495111 Date: 2026.05.22 12:35:53 -04'00'	

PATIENT IDENTIFICATION			
PATIENT'S NAME (Last, First, Middle Initial)	(Male/Female)	DATE OF BIRTH	SSN OR IDENTIFICATION NO.

Defense Occupational and Environmental Health Readiness System (DOEHRS) (August 2012)

CUI HLTH//TSCA

Controlled By: DHA Categories: HLTH//TSCA
LDC: FEDCON, DOD Only
POC:DE TATA, T

Page 2 of 2

Annual Review Log:
Scott Morrill / 30Oct2026 / DSN: 312-857-6298

Trainee Name	Employee ID / Last 4	Date Trained	Trainer / Supervisor Signature